Krish Kochar

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Education

University of Waterloo Sep 2022 - Apr 2027

Bachelor of Computer Science (AI Specialization) ***GPA: 4.0***

Selected Coursework: Algorithms, Operating Systems, Concurrency, Distributed Systems, Machine Learning, Computer Security

Experience

AWS Transactional Services Sep 2026 - Dec 2026

Incoming Software Engineering Intern

- On the Control Plane team for the Journal Service, powering consistency & durability in DynamoDB & S3

Atlassian May 2026 - Aug 2026

Incoming Software Engineering Intern

Super.com May 2025 - Aug 2025

Software Engineering Intern

- Engineered **Go** backend services for booking, payment, & refund workflows, achieving **<50ms p95** service latency
- Designed an async webhook-based refund dispute pipeline with idempotency guarantees, ensuring correct semantics under retries, duplicate events, & partial failures, reducing **resolution turnaround time by 30%**
- Integrated a third-party fraud service into the payments API, reducing **fraudulent bookings by 25%**
- Drove production triage, resolving **5+** critical incidents & introducing debugging workflows that reduced **MTTR by 50%**

OTTO by Rockwell Automation Jan 2025 – Apr 2025

Software Developer Intern

- Developed a multi-fleet interoperability system in **C++**, **Python**, & **ROS2**, improving coordination across **500+** robots
- Redesigned **networking architecture** by migrating from CycloneDDS to Zenoh, reducing latency by **25%** & increasing throughput by **30%**, validated with benchmarks & a production-ready **PoC** for fleet deployment
- Iteratively tuned **QoS** settings to reduce packet loss in stress tests, boosting E2E communication across distributed nodes
- Replaced polling with an **event-driven ROS2** topic for real-time robot tracking, reducing CPU usage by **20%**

Ford May 2024 - Aug 2024

Platform Software Developer Intern

- Deployed a real-time Vision Compute Service in **C++** with **OpenVX** & **OpenCV**, accelerating sensor data processing & optimizing multi-threaded workloads for high-performance execution on embedded hardware
- Integrated **Gcov** into **Jenkins** CI pipelines, enabling **80%** functional test coverage visibility & unblocking QA workflows
- Debugged & resolved **critical memory defects** with Valgrind, eliminating leaks & stabilizing production firmware

Publicis Sapient May 2023 - Aug 2023

Software Engineering Intern

- Optimized API latency by **30%** via redesigned data-fetching paths & **GraphQL** aggregation to reduce network overhead

Research

University of Waterloo Jan 2026 - Present

Undergraduate Research Assistant, Prof. Sujaya Maiyya

- Designing a **privacy-preserving oblivious vectorDB** for secure, efficient similarity search that hides access patterns
- Building an oblivious **HNSW similarity search** in **C++** within TEEs to minimize query latency & side-channel leakages

University of Waterloo & Vector Institute Sep 2024 - Dec 2024

Undergraduate Research Assistant, Prof. Xi He

- Generated a synthetic dataset with **<10%** divergence from real data, using a ClavaDDPM **diffusion** pipeline in **Python**
- Benchmarked data privacy via black-box **Membership Inference Attacks** to assess the risks of sharing synthetic data

Technical Skills

Languages: C, C++, Python, Go, Rust, Java, SQL, TypeScript, JavaScript, Scala, Bash

Systems & Tools: Docker, Kubernetes, Helm, Jenkins, Git, Redis, Datadog, Google Test

Frameworks & Libraries: ROS2, OpenCV, OpenVX, FastAPI, React, Node.js, PyTorch, MapReduce, Spark